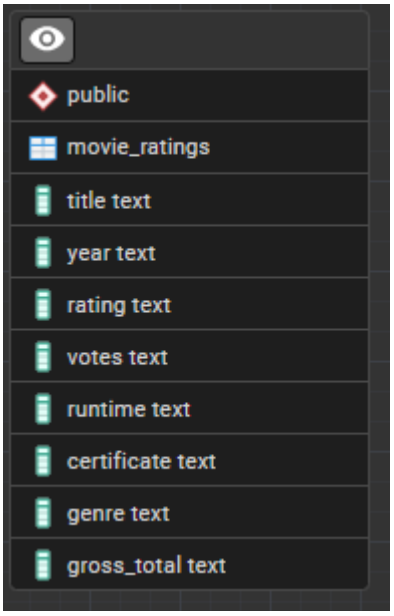
Movie Ratings: Queries & Insights

Queries

-- ERD



-- Q1 Top 10 Highest-Rated Movies

```
SELECT title, rating
FROM movie_ratings
ORDER BY rating DESC
LIMIT 10
```

	title text	rating text
1	The Shawshank Redemption	9.3
2	The Godfather	9.2
3	The Dark Knight	9.1
4	The Godfather Part II	9.0
5	The Lord of the Rings: The Return of the King	9.0
6	12 Angry Men	9.0
7	Schindler's List	9.0
8	The Lord of the Rings: The Fellowship of the...	8.9
9	Pulp Fiction	8.8
10	Il buono, il brutto, il cattivo	8.8

-- Q2 Top 10 Most-Voted Movies

```
SELECT
    title,
    rating::real AS rating_value,
    REPLACE(votes, ',', '')::bigint AS votes_numeric
FROM movie_ratings
ORDER BY votes_numeric DESC
LIMIT 10;
```

	title text	rating_value real	votes_numeric bigint
1	The Shawshank Redemption	9.3	3168010
2	The Dark Knight	9.1	3145646
3	Inception	8.8	2796318
4	Fight Club	8.8	2585212
5	Interstellar	8.7	2496091
6	Forrest Gump	8.8	2479948
7	Pulp Fiction	8.8	2421067
8	The Matrix	8.7	2233701
9	The Godfather	9.2	2212884
10	The Lord of the Rings: The Fellowship of the...	8.9	2193491

-- Q3 Movie Count by Certificate

```
SELECT
    certificate,
    COUNT(*) AS movie_count
FROM movie_ratings
GROUP BY certificate
ORDER BY movie_count DESC;
```

	certificate text	movie_count bigint
1	R	102
2	Unrated	57
3	PG	38
4	PG-13	34
5	G	18
6	NC-17	1

-- Q4 Average Movie Ratings by Certificate

```
SELECT certificate,
       ROUND(AVG(rating::real)::numeric, 2) AS avg_rating
FROM movie_ratings
```

```
GROUP BY certificate
ORDER BY avg_rating DESC;
```

	certificate text	avg_rating numeric
1	PG-13	8.38
2	R	8.33
3	NC-17	8.30
4	PG	8.28
5	Unrated	8.27
6	G	8.26

-- Q5 Top-Rated Movie per Certificate

```
WITH ranked AS (
  SELECT
    Title,
    Certificate,
    Rating,
    Votes,
    RANK() OVER (PARTITION BY Certificate ORDER BY Rating DESC) AS
rank
  FROM movie_ratings
)

SELECT Title, Certificate, Rating, Votes
FROM ranked
WHERE rank = 1
ORDER BY rating DESC;
```

	title text	certificate text	rating text	votes text
1	The Shawshank Redemption	R	9.3	3,168,010
2	The Dark Knight	PG-13	9.1	3,145,646
3	12 Angry Men	Unrated	9.0	977,359
4	Star Wars: Episode V - The Empire Strikes B...	PG	8.7	1,496,595
5	Modern Times	G	8.5	284,271
6	The Lion King	G	8.5	1,251,328
7	City Lights	G	8.5	215,684
8	Requiem for a Dream	NC-17	8.3	968,756

```sql
-- Q6 Top-Rated Movie per Genre
WITH genre_split AS (
  SELECT
    title,
    rating,
    votes,
    TRIM(unnest(string_to_array(genre, ','))) AS genre
  FROM movie_ratings
),

ranked AS (
  SELECT *,
         RANK() OVER (PARTITION BY genre ORDER BY rating DESC) AS
rank
  FROM genre_split
)

SELECT title, genre, rating, votes
FROM ranked
WHERE rank = 1
ORDER BY genre;
```

	title text	genre text	rating text	votes text
1	The Dark Knight	Action	9.1	3,145,646
2	The Lord of the Rings: The Return of the King	Adventure	9.0	2,152,126
3	Sen to Chihiro no kamikakushi	Animation	8.6	959,384
4	Schindler's List	Biography	9.0	1,577,558
5	La vita è bella	Comedy	8.6	808,265
6	The Godfather	Crime	9.2	2,212,884
7	The Shawshank Redemption	Drama	9.3	3,168,010
8	It's a Wonderful Life	Family	8.6	552,664
9	Sen to Chihiro no kamikakushi	Family	8.6	959,384
10	The Lord of the Rings: The Return of the King	Fantasy	9.0	2,152,126
11	Sunset Blvd.	Film-Noir	8.4	260,009
12	Schindler's List	History	9.0	1,577,558
13	The Silence of the Lambs	Horror	8.6	1,711,212
14	The Pianist	Music	8.5	1,006,347
15	Whiplash	Music	8.5	1,143,440
16	Singin' in the Rain	Musical	8.3	281,252
17	Se7en	Mystery	8.6	2,006,146
18	Seppuku	Mystery	8.6	89,994

-- Q7 Movies Categorized by Runtime

```
SELECT title,rating, runtime, genre,
      CASE
        WHEN runtime::integer <= 60 THEN 'Under 1hr'
        WHEN runtime::integer <= 120 THEN 'Under 2hrs'
        WHEN runtime::integer <= 180 THEN 'Under 3hrs'
        WHEN runtime::integer <= 240 THEN 'Under 4hrs'
        WHEN runtime::integer <= 300 THEN 'Under 5hrs'
    ELSE 'More than 5hrs'
END AS runtime_category
FROM movie_ratings
ORDER BY runtime::integer DESC;
```

	title text	rating text	runtime text	genre text	runtime_category text
1	La meglio gioventù	8.4	374	Drama, Romance	More than 5hrs
2	Gangs of Wasseypur	8.2	321	Action, Comedy, Crime	More than 5hrs
3	Kill Bill: The Whole Bloody Affair	8.8	253	Action, Crime, Thriller	Under 5hrs
4	Gone with the Wind	8.2	238	Drama, Romance, War	Under 4hrs
5	C'era una volta in America	8.3	229	Crime, Drama	Under 4hrs
6	Lawrence of Arabia	8.3	227	Adventure, Biography, ...	Under 4hrs
7	Ben-Hur	8.1	212	Adventure, Drama	Under 4hrs

-- Q8 Top Movie per Year Ranked Overall by Rating

```
WITH top_per_year AS (
    SELECT title, year, rating, runtime, genre,
           ROW_NUMBER() OVER (
           PARTITION BY year
           ORDER BY rating::real DESC
           ) AS rank_per_year
    FROM movie_ratings
)

SELECT title, year, rating, runtime, genre, ROW_NUMBER() OVER
(ORDER BY rating::real DESC) AS overall_rank
FROM top_per_year
WHERE rank_per_year = 1
ORDER BY year DESC
```

	title text	year text	rating text	runtime text	genre text	overall_rank bigint
1	Kill Bill: The Whole Bloody Affair	2025	8.8	253	Action, Crime, Thriller	8
2	Dune: Part Two	2024	8.4	166	Action, Adventure, Dra...	51
3	12th Fail	2023	8.7	147	Biography, Drama	16
4	Top Gun: Maverick	2022	8.2	130	Action, Drama	73
5	Jai Bhim	2021	8.6	164	Crime, Drama	24
6	Hamilton	2020	8.3	160	Biography, Drama, His...	54
7	Gisaengchung	2019	8.5	132	Drama, Thriller	41
8	Capharnaüm	2018	8.4	126	Drama	42
9	Kimi no na wa.	2017	8.4	106	Animation, Drama, Fa...	52

```sql
-- Q9 Movies Categorized by Box Office Performance
WITH movies_numeric AS (
SELECT *,
            CASE
                WHEN gross_total LIKE '%B' THEN
                     LEFT(REPLACE(gross_total, '$', ''),
LENGTH(REPLACE(gross_total, '$', '')) - 1)::numeric * 1000000000
                WHEN gross_total LIKE '%M' THEN
                     LEFT(REPLACE(gross_total, '$', ''),
LENGTH(REPLACE(gross_total, '$', '')) - 1)::numeric * 1000000
                ELSE
                     REPLACE(gross_total, '$', '')::numeric
            END AS gross_numeric
    FROM movie_ratings
)

SELECT
    title,year, rating, gross_total,
    CASE
        WHEN gross_numeric IS NULL THEN 'Unknown'
        WHEN gross_numeric >= 1000000000 THEN 'Blockbuster'
        WHEN gross_numeric >= 500000000 THEN 'Hit'
        WHEN gross_numeric >= 100000000 THEN 'Average Hit'
        ELSE 'Flop'
    END AS box_office_category
FROM movies_numeric
ORDER BY gross_numeric DESC;
```

	title text	year text	rating text	gross_total text	box_office_category text
1	Hababam Sinifi: Sinifta Kaldi	1975	8.5	[null]	Unknown
2	Drishyam	2015	8.2	[null]	Unknown
3	Jai Bhim	2021	8.6	[null]	Unknown
4	Klaus	2019	8.2	[null]	Unknown
5	Avengers: Endgame	2019	8.4	$2799.44M	Blockbuster
6	Avengers: Infinity War	2018	8.4	$2052.42M	Blockbuster
7	Spider-Man: No Way Home	2021	8.1	$1921.43M	Blockbuster
8	Top Gun: Maverick	2022	8.2	$1495.70M	Blockbuster
9	Harry Potter and the Deathly Hallows: Part 2	2011	8.1	$1342.94M	Blockbuster
10	The Lord of the Rings: The Return of the King	2003	9.0	$1148.97M	Blockbuster
11	Jurassic Park	1993	8.2	$1103.11M	Blockbuster
12	The Dark Knight Rises	2012	8.4	$1085.43M	Blockbuster
13	Joker	2019	8.3	$1078.96M	Blockbuster
14	Toy Story 3	2010	8.3	$1067.32M	Blockbuster
15	The Dark Knight	2008	9.1	$1008.48M	Blockbuster
16	The Lion King	1994	8.5	$979.16M	Hit
17	Oppenheimer	2023	8.2	$975.81M	Hit
18	The Lord of the Rings: The Two Towers	2002	8.8	$944.74M	Hit
19	Finding Nemo	2003	8.2	$941.64M	Hit
20	The Lord of the Rings: The Fellowship of the Ring	2001	8.9	$897.00M	Hit
21	Inside Out	2015	8.1	$859.08M	Hit

-- Q10 Yearly Top Movie Ratings and Changes

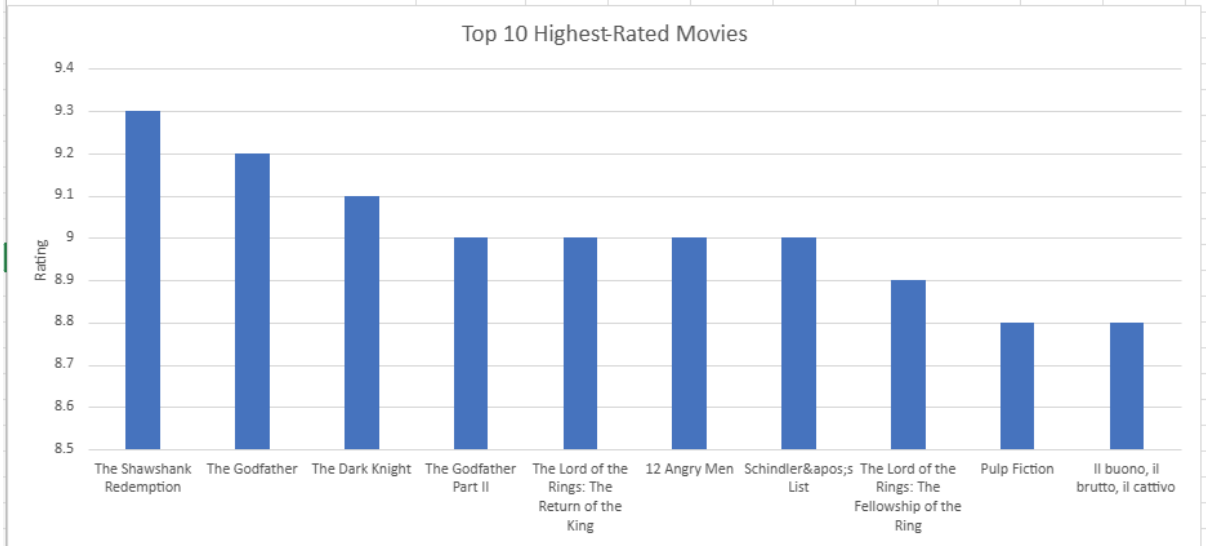
```
WITH top_per_year AS (
    SELECT *,
           ROW_NUMBER() OVER (
               PARTITION BY year
               ORDER BY rating::real DESC
           ) AS rank_per_year
    FROM movie_ratings
)

SELECT
    title,
    year,
    rating::real,
    LAG(rating::real) OVER (ORDER BY year) AS prev_year_rating,
    rating::real - LAG(rating::real) OVER (ORDER BY year) AS
rating_change
FROM top_per_year
WHERE rank_per_year = 1
ORDER BY year;
```

Insights

Query 1:

title	rating
The Shawshank Redemption	9.3
The Godfather	9.2
The Dark Knight	9.1
The Godfather Part II	9
The Lord of the Rings: The Return of the King	9
12 Angry Men	9
Schindler's List	9
The Lord of the Rings: The Fellowship of the Ring	8.9
Pulp Fiction	8.8
Il buono, il brutto, il cattivo	8.8



Observations: The bar chart shows that all films have very similar ratings, ranging narrowly from 8.8 to 9.3, indicating consistently high quality across the dataset. The Shawshank Redemption stands slightly above the rest as the highest-rated film, but the differences between most movies are minimal. Several films share the same or nearly identical ratings, creating visible clusters of bars with similar heights.

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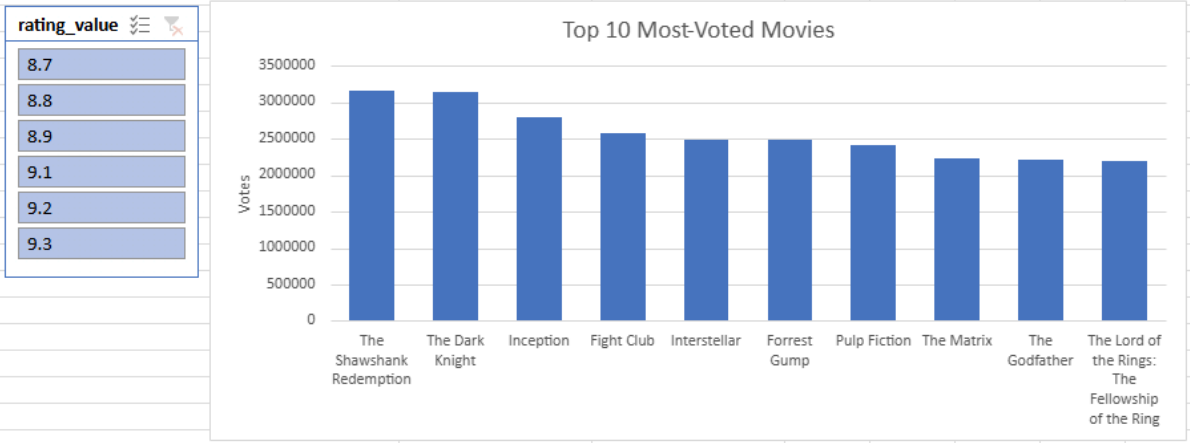
Query 2:

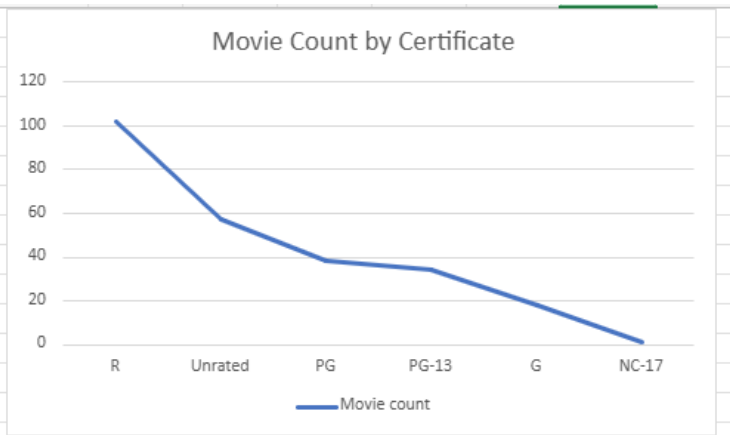
title	Sum of votes_numeric
The Shawshank Redemption	3168010
The Dark Knight	3145646
Inception	2796318
Fight Club	2585212
Interstellar	2496091
Forrest Gump	2479948
Pulp Fiction	2421067
The Matrix	2233701
The Godfather	2212884
The Lord of the Rings: The Fellowship of the Ring	2193491
Grand Total	**25732368**



Observations: This chart shows that some movies are not only highly rated but also super popular, with The Shawshank Redemption and The Dark Knight getting the most votes. Even films with slightly lower ratings, like Inception and Fight Club, still have huge numbers of viewers. Overall, it suggests that while ratings are similar, some movies simply reach a much bigger audience.

Query 3:

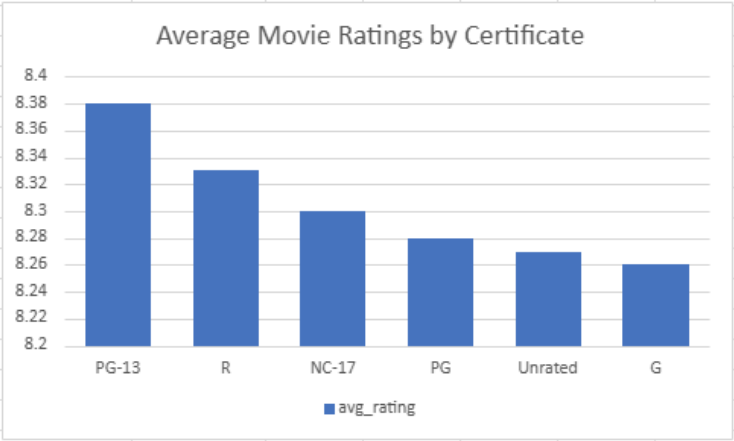
certificate	movie_count
R	102
Unrated	57
PG	38
PG-13	34
G	18
NC-17	1



Observations: Out of the 250 top movies in the dataset, the majority are rated R, showing a strong presence of more mature content. Categories like Unrated, PG, and PG-13 have a moderate number of films, while G-rated movies are relatively few. NC-17 is almost nonexistent, with only one film, highlighting how rare that certification is among top-ranked movies.

Query 4:

certificate	avg_rating
PG-13	8.38
R	8.33
NC-17	8.3
PG	8.28
Unrated	8.27
G	8.26



Observations: The chart shows that average ratings are very close across all certificate categories, ranging only from about 8.26 to 8.38. PG-13 has the highest average rating, but only slightly above R and NC-17, indicating no major difference in quality between categories. Overall, film ratings appear consistent regardless of certification, with only minimal variation.

Query 5:

Single value visualization:

	title text	certificate text	rating text	votes text
1	The Shawshank Redemption	R	9.3	3,168,010
2	The Dark Knight	PG-13	9.1	3,145,646
3	12 Angry Men	Unrated	9.0	977,359
4	Star Wars: Episode V - The Empire Strikes B...	PG	8.7	1,496,595
5	Modern Times	G	8.5	284,271
6	The Lion King	G	8.5	1,251,328
7	City Lights	G	8.5	215,684
8	Requiem for a Dream	NC-17	8.3	968,756

Observations: The data shows that top-rated films span all certificate categories, indicating high quality isn't limited by certification. The Shawshank Redemption leads R in both rating and votes, while The Dark Knight (PG-13) and The Lion King (G) also perform strongly. Even less common categories like NC-17 feature acclaimed films such as Requiem for a Dream.

Query 6:

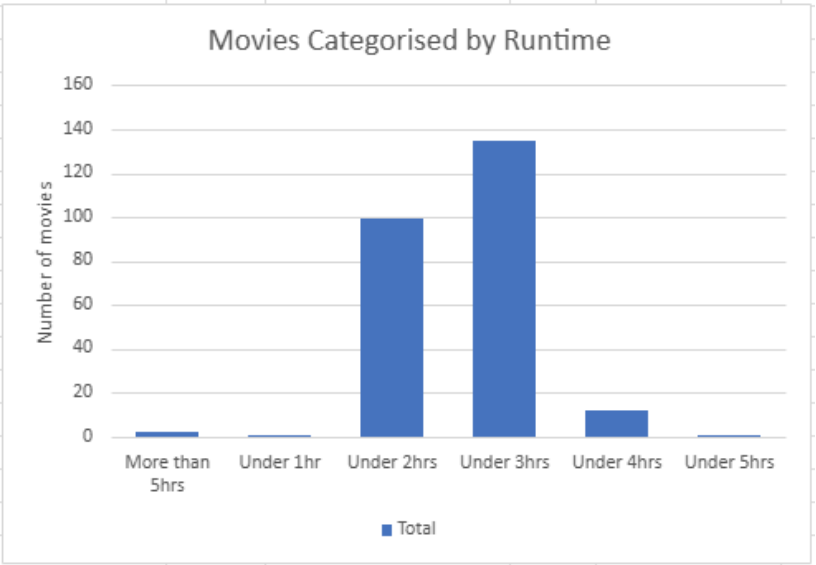
Single value visualization:

	title text	genre text	rating text	votes text
1	The Dark Knight	Action	9.1	3,145,646
2	The Lord of the Rings: The Return of the King	Adventure	9.0	2,152,126
3	Sen to Chihiro no kamikakushi	Animation	8.6	959,384
4	Schindler's List	Biography	9.0	1,577,558
5	La vita è bella	Comedy	8.6	808,265
6	The Godfather	Crime	9.2	2,212,884
7	The Shawshank Redemption	Drama	9.3	3,168,010
8	It's a Wonderful Life	Family	8.6	552,664
9	Sen to Chihiro no kamikakushi	Family	8.6	959,384
10	The Lord of the Rings: The Return of the King	Fantasy	9.0	2,152,126
11	Sunset Blvd.	Film-Noir	8.4	260,009
12	Schindler's List	History	9.0	1,577,558
13	The Silence of the Lambs	Horror	8.6	1,711,212
14	The Pianist	Music	8.5	1,006,347
15	Whiplash	Music	8.5	1,143,440
16	Singin' in the Rain	Musical	8.3	281,252
17	Se7en	Mystery	8.6	2,006,146
18	Seppuku	Mystery	8.6	89,994

Observations: The chart highlights the top movie in each genre, showing the most notable title for every category.

Query 7:

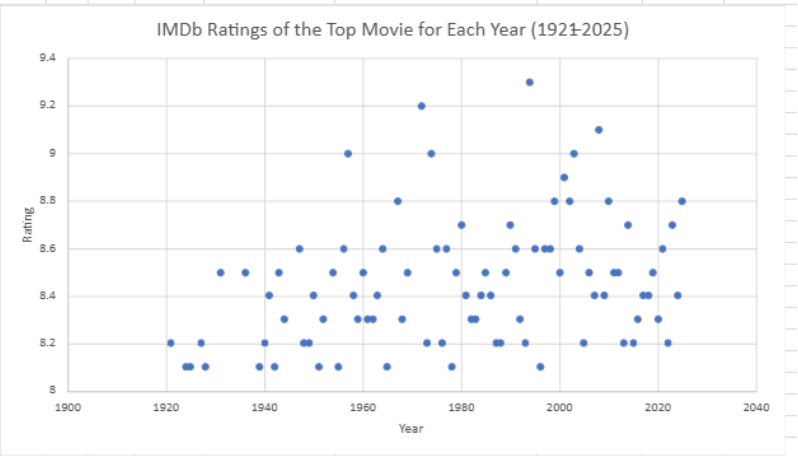
runtime_category	Count of title
More than 5hrs	2
Under 1hr	1
Under 2hrs	99
Under 3hrs	135
Under 4hrs	12
Under 5hrs	1
Grand Total	**250**



Observations: The chart shows the distribution of movies by runtime among the 250 top titles. Most films fall under 2–3 hours, with 99 under 2 hours and 135 under 3 hours. Very long movies are rare, with only 2 exceeding 5 hours, while extremely short films are even less common, with just 1 under 1 hour.

Query 8:

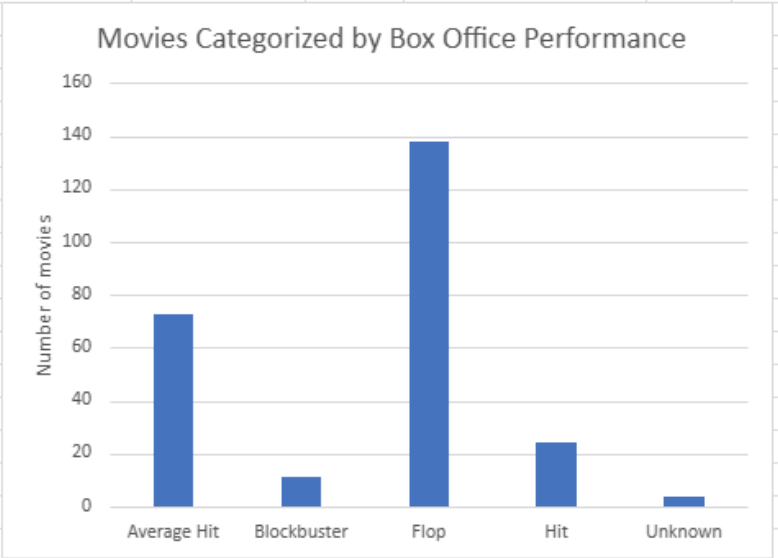
year	title	rating	overall_rank
2025	Kill Bill: The Whole Bloody Affair	8.8	8
2024	Dune: Part Two	8.4	51
2023	12th Fail	8.7	16
2022	Top Gun: Maverick	8.2	73
2021	Jai Bhim	8.6	24
2020	Hamilton	8.3	54
2019	Gisaengchung	8.5	41
2018	CapharnaÃ¼m	8.4	42
2017	Kimi no na wa.	8.4	52
2016	Dangal	8.3	55
2015	Drishyam	8.2	76
2014	Interstellar	8.7	14
2013	Prisoners	8.2	75
2012	Django Unchained	8.5	40
2011	Intouchables	8.5	39
2010	Inception	8.8	9
2009	3 Idiots	8.4	50
2008	The Dark Knight	9.1	3
2007	Das Leben der Anderen	8.4	49
2006	The Departed	8.5	38
2005	Der Untergang	8.2	72
2004	Cidade de Deus	8.6	19
2003	The Lord of the Rings: The Return (	9	4



Observations: Over the years, top-rated films have consistently earned high ratings, without any clear pattern of improvement or decline. Classics like *The Godfather* (1972) and *12 Angry Men* (1957) still rank among the best, alongside more recent hits like *The Dark Knight* (2008) and *Kill Bill: The Whole Bloody Affair* (2025). This suggests that great movies can come from any era, and quality depends more on the film itself than when it was released.

Query 9:

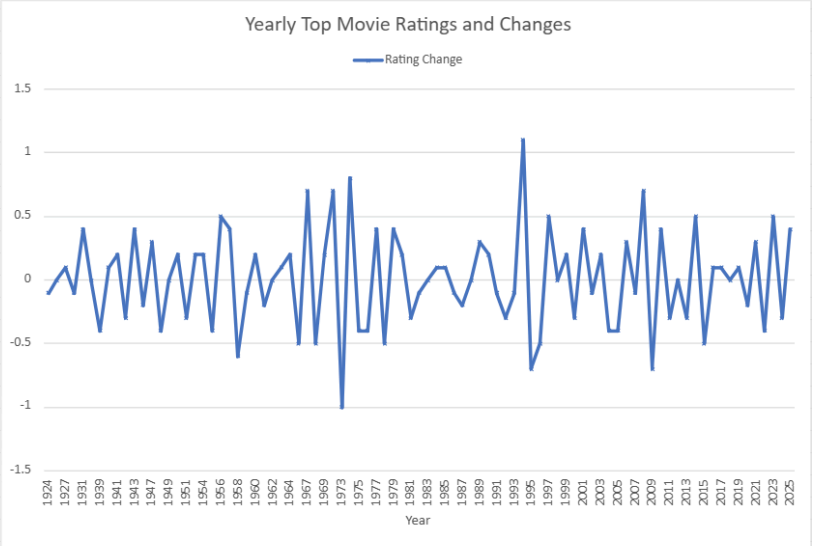
box_office_category	Count of title
Average Hit	73
Blockbuster	11
Flop	138
Hit	24
Unknown	4
Grand Total	**250**



Observations: The chart shows the distribution of movies by box office performance among the 250 top titles. Most films are classified as Flops (138), while Average Hits make up a significant portion (73). Fewer movies achieve Hit (24) or Blockbuster (11) status, and a small number (4) have Unknown box office results, highlighting that commercial success does not always align with critical acclaim.

Query 10:

year	rating	prev_year_rating	rating_change
1921	8.2	NULL	NULL
1924	8.1	8.2	-0.09999943
1925	8.1	8.1	0
1927	8.2	8.1	0.09999943
1928	8.1	8.2	-0.09999943
1931	8.5	8.1	0.39999962
1936	8.5	8.5	0
1939	8.1	8.5	-0.39999962
1940	8.2	8.1	0.09999943
1941	8.4	8.2	0.19999981
1942	8.1	8.4	-0.29999924
1943	8.5	8.1	0.39999962
1944	8.3	8.5	-0.19999981
1947	8.6	8.3	0.3000002
1948	8.2	8.6	-0.40000057
1949	8.2	8.2	0
1950	8.4	8.2	0.19999981
1951	8.1	8.4	-0.29999924
1952	8.3	8.1	0.19999981
1954	8.5	8.3	0.19999981
1955	8.1	8.5	-0.39999962
1956	8.6	8.1	0.5
1957	9	8.6	0.39999962
1958	8.4	9	-0.6000004
1959	8.3	8.4	-0.09999943
1960	8.5	8.3	0.19999981



Observations: Looking at how ratings change from year to year, there's no clear upward or downward trend—some years see big jumps, while others dip a bit. For example, The Shawshank Redemption (1994) jumped by +1.1, and The Godfather Part II (1974) went up by +0.8, while The Sting (1973) dropped by -1.0, and Se7en (1995) went down by -0.7. Overall, ratings seem to fluctuate year by year, showing that each movie's reception is unique rather than part of a steady trend.